

Dancing Stick Figures: An Introductory Dataset for Training Video Generation Models

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[Dataset](#) · [Code](#) · [Models](#) · [Colab](#)

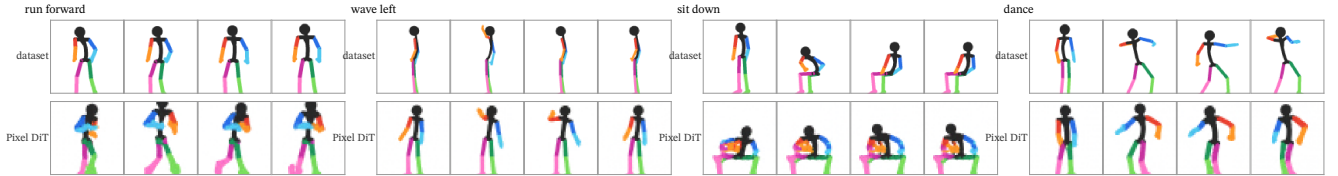


Figure 1. Released dataset clips and outputs from the 30k-step image-initialized Pixel DiT for locomotion, gesture, transition, and dance prompts. Each block shows frames 0/21/42/63 from a released clip and an independently seeded generation (50 Euler steps, guidance 3).

Abstract

*Training a video-generation model from scratch is hard for reasons that precede model design. The feedback loop is long: a failure that appears only after a training run can make each attempted fix another run. The data are hard to reach: the corpora and recipes behind strong models are large, heterogeneous, and often unreleased. And scoring is blunt: open-ended generation has no single correct output, and an aggregate score does not by itself establish whether a sample succeeds or which property failed. **Dancing Stick Figures** is a synthetic video dataset built against these three obstacles. For iteration speed, its 64^2 , 64-frame reference task is sized for practical repeated training on a single workstation GPU. For accessibility, the release is a 0.79-GB training tier of 4,020 video clips—1,340 six-second source motions, each rendered from three cameras by a deterministic dataset-generation harness—with checkpoints and a Colab workflow that reruns the reference training pipeline at reduced budget on a 16 GB Tesla T4. For scoring, every frame retains its generating state (ARDY cskel27 joint positions, camera, body parameters, and source motion) and per-pixel depth, surface normals, and part labels. These annotations support dataset-specific metrics for visible topology and part-wise motion; corruptions expose their sensitivities and blind spots.*

1. Introduction

Training a video-generation model from scratch takes a heavy toll on the wallet. The reference recipes assume GPU servers, and renting them turns every experiment into a billing decision—a poor mode of study. Study wants a small, annotation-rich dataset one can play with: download it, inspect it, train on it repeatedly, and change one thing at a time. Public datasets of real footage exist at every scale, but the ones rich enough to train a video model are typically too large for repeated from-scratch training, and collected footage records nothing about the true scene behind each frame. What is missing is a dataset purpose-built for research: small enough to fit one GPU, fast enough to see an end-to-end result within a day, and procedural, so that a researcher can tweak the generator and regenerate the data—and so the generating parameters themselves can seed score functions. We built Dancing Stick Figures for that role. Three barriers shape its design:

- **Iteration cost.** The complete training path for frontier systems such as Seedance and MiniMax H3 [9, 10] runs through large corpora, heavy compute, pretrained codecs, and often proprietary artifacts; controlled retraining there is impractical. A failure that appears only after training—for example, motion that becomes static late in a clip—can make each attempted diagnosis another expensive run.
- **Accessibility.** Public datasets, training code, pretrained

components, and benchmarks are available, but rarely as one documented, runnable workflow. A researcher must still choose compatible components, reconstruct preprocessing and training details, and determine what results to expect. This integration burden makes it difficult to begin with a small experiment and modify it with confidence.

- **Diagnosis.** Open-ended generation has no single correct output, while failures differ: motion may become static, flicker, or lose smoothness; subject appearance may drift; and the action may miss the prompt. These correspond to distinct evaluation dimensions such as dynamic degree, temporal flickering, motion smoothness, subject consistency, and prompt adherence [18]; one aggregate score cannot identify which property failed.

These barriers lead to three corresponding design choices: a compact reference task for repeated training, a connected release that runs end to end, and procedural data whose generating state supports dataset-specific diagnosis.

The procedural design exposes variables that are usually unavailable in collected video. Each rendered frame is linked to its source motion, rig and body, camera, renderer, depth, surface normals, and part labels. We can change one variable at a time, rerender the clip, and observe which diagnostics react. Claims from these interventions are limited to the recorded variables and controlled changes tested here.

Can a researcher train reference models from scratch on a single workstation GPU and then iterate on them through short, controlled cycles? Each cycle should expose a named failure, change one mechanism, and rerun the same diagnostic. The release supports that workflow with a connected end-to-end route, an image checkpoint used to initialize the video model’s spatial weights, runs short enough to repeat and vary, and diagnostics with known blind spots.

Dancing Stick Figures connects three operations: regenerate the dataset from its recorded source parameters, train a 64-frame prompt-conditioned video generator, and evaluate its outputs with validated dataset-specific metrics.

The release contributes:

1. a label-rich video dataset released at 128² and in a 0.79-GB 64² tier, with the motion, camera, and body parameters behind every frame, and a deterministic dataset-generation harness;
2. a seed-disjoint evaluation protocol and dataset-specific topology and motion metrics, validated with a controlled corruption suite; and
3. image and video trainers, checkpoints, and a reduced-budget Colab workflow that reruns the reference training pipeline end to end.

2. Related Work

Controlled synthetic data. Moving MNIST [1] made temporal prediction cheap to study, while Kubric and MOVi [2] programmatically render scenes with dense

ground truth for scene understanding. SURREAL [3] and BEDLAM [4] extend synthetic rendering to articulated people at a scale and realism aimed at perception. Dancing Stick Figures focuses on text-conditioned videos of a single stick figure. The small, controlled domain makes end-to-end retraining affordable and lets us test the failure diagnostics with known corruptions.

Video training data and motion sources. HumanVid [5] and MiraData [6] broaden the coverage of video training data, while AMASS [7] and HumanML3D [8] release motion representations. Dancing Stick Figures connects these layers in one compact release: text-conditioned source trajectories are rendered into video while preserving the correspondence among prompts, joints, camera and body parameters, limb labels, and pixels. ARDY [11] supplies the upstream 27-joint trajectories; it is not evaluated by our video diagnostics. This pairing lets the released pixels train a video model, the known limb structure define dataset-specific measurements, and controlled corruptions test what those measurements detect.

Evaluation of generated video. FVD [12] is widely reported as a distributional comparison in video-generation benchmarks [16]. It is not a diagnostic of whether an individual video succeeds: its I3D features have a documented content bias and can be under-sensitive to temporal corruption [13]. Our controlled study uses FVD as a conventional coarse reference and tests that limitation directly in this domain. VBench [18] demonstrates the value of separating video quality into diagnostic dimensions. Our setting adds recorded generating state, which lets us apply known corruptions and test which dataset-specific signals respond.

3. Dataset and generation pipeline

We generated one motion per seed 0–9 for each of 143 prompts in six groups (dance, gesture, locomotion, transition, idle, sport) using NVIDIA ARDY [11]. Its Core model returns batchable 20-fps trajectories on a fixed 27-joint skeleton, matching the harness input without skeleton conversion. Each trajectory retains its prompt association, while body shape, camera, and raster appearance are applied later under deterministic control. Each motion is six seconds at 20 fps.

ARDY supplies only the source motion; the reference models in this paper generate rendered video and are evaluated with the video diagnostics. The released motion table lets users rebuild or vary the rendered dataset without running ARDY. For additional source-motion generation, we use ARDY v0.2.0. The released motion table remains the canonical input for reproducing the published dataset. The resulting motion distribution and prompt response inherit

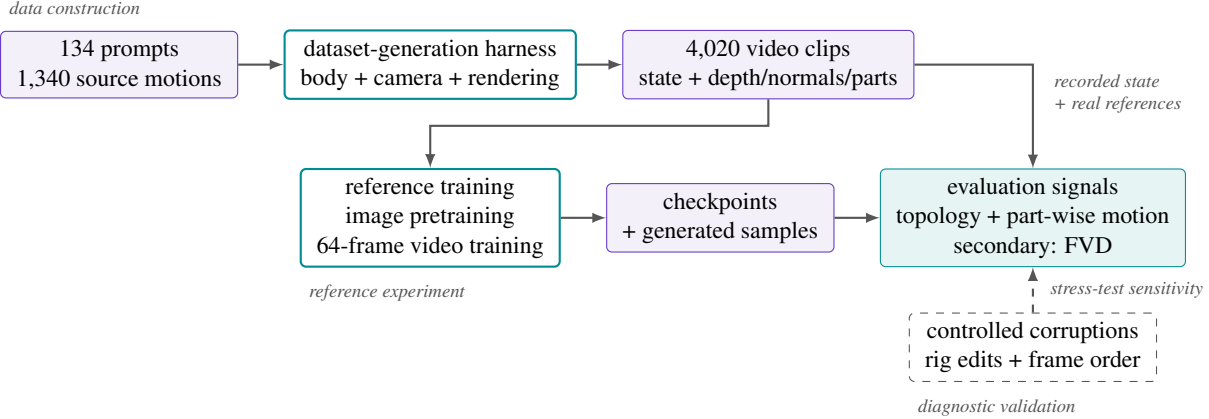


Figure 2. System map of the release. The released dataset is an artifact of the dataset-generation harness applied to the released prompts and source motions. Reference training consumes the dataset and produces model artifacts: checkpoints and generated samples. Recorded state and real references from the dataset, together with generated samples, feed the evaluation signals; controlled corruptions validate what those signals detect and miss.

Table 1. Dataset configurations and storage footprint. A motion is rendered from three views; `frames` and `mini` store one record per rendered frame, `motion` one record per source motion.

Configuration	Records	Resolution	Size
<code>frames</code>	482,400	128 ²	4.4 GB
<code>mini</code>	482,400	64 ²	0.79 GB
<code>motion</code>	1,340	27 joints	0.33 GB

Table 2. Seed-disjoint splits. Each split contains all 134 prompts; source motions are disjoint by seed.

Split	Prompts	Motions	Videos	Frames
Train	134	1,072	3,216	385,920
Validation	134	134	402	48,240
Test	134	134	402	48,240
All	134	1,340	4,020	482,400

ARDY’s capabilities and biases. Licensing and provenance are detailed in §6.

A capsule renderer with $4\times$ supersampling emits three deterministic orthographic views. The release records 143 candidate prompts; nine are marked for exclusion with per-prompt reasons, leaving 134 in the dataset. Selection is at prompt level, so all seeds of retained prompts remain. Table 1 summarises the release. Seeds 0–7 train, seed 8 validates, and seed 9 tests: every released prompt appears in all three splits, and no source motion crosses a split boundary (Table 2). Because selection considered realisations across seeds, held-out evaluation measures unseen motion realisations of known, selected prompts, not unseen-prompt generalisation.

The generator samples limb length, scale, stroke, and

camera yaw and pitch deterministically from `clip_id`. It stores 3D and projected joints, visibility, camera and body parameters, RGBA, 16-bit depth, camera-space normals, 27-part segmentation, and the raw motion (Figure 3); the dataset card is the schema reference. Released rendering parameters span 50–58 pixels/m, 3–5-pixel strokes, -3° – 10° camera pitch, and .92–1.08 per-bone scale multipliers. The six groups contain 33 dance, 29 gesture, 21 locomotion, 13 transition, 10 idle, and 28 sport prompts. The builder records two motion-quality flags: `frozen` (mean joint speed below 0.02 m/s) and `levitation` (root ever above 1.6 m). Of 1,340 source motions, 58 are marked frozen (41 from the idle group) and one levitation; flagged samples stay in the release so filtering choices are explicit. Across 4,020 rendered clips, the median foreground fraction is .059 (5th–95th percentiles .043–.077). After deduplicating views by retaining camera 0 once per source motion, median centroid speed is .269 pixels/frame (.058–.635) and summed projected-rig part angular path is 24.75 rad (2.37–94.99). The largest train–validation or train–test Kolmogorov–Smirnov distance across these three quantities is .106.

The released configurations are two resolutions of one dataset: 128² keeps the most detail, and the 0.79-GB 64² tier is the primary training tier; a derived 32² cache view supports smoke tests. Reported structural results use 64², where the thin colored limbs remain measurable. The motion release supports rerendering the released motions into the full visual dataset without ARDY. §6 describes the rebuild and its verification.

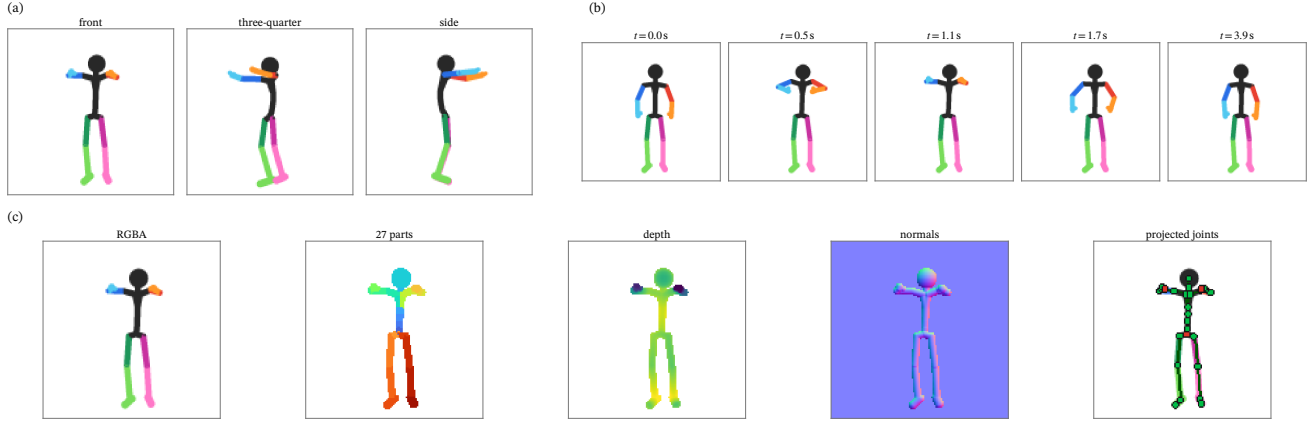


Figure 3. One released motion—A person waves goodbye with both hands.—viewed through the dataset. (a) The same representative pose from front, three-quarter, and side cameras. (b) Five frames at equal increments of pose-space motion, so an idle tail cannot dominate the strip. (c) The same frame with part segmentation, depth, camera-space normals, and projected joints. Each record also stores its prompt, source-motion identifier, seed, camera, and body parameters.

4. Reference benchmark and evaluation

Generation setup. We use the dataset to define a compact reference benchmark for prompt-conditioned video generation. Each public clip has 120 frames at 20 fps. Every benchmark and diagnostic experiment in this paper uses the fixed first 64-frame window at native 20 fps, matching the reference-model output. The reference models train on and generate the fixed first 64 frames of each clip (3.2 seconds). The window rows in Table 3 compare generated 64-frame windows with real windows drawn under the same manifest rule.

Structural diagnostics. Each arm and leg uses a fixed pair of colors. The evaluator converts a frame into eight limb-color masks and reports:

- **Topology violation rate (TVR):** limb colors missing or split into multiple components;
- **Limb-identity error (LIE):** expected color adjacencies that are absent;
- **Color-purity error (CPE):** foreground pixels outside the rendering palette;
- **Foreground fraction:** pixels occupied by the figure; and
- **Clean rate:** frames with both TVR and LIE at zero.

Implementation uses 8-connected components (a 3-by-3 neighborhood) for TVR. For LIE, an expected color pair is adjacent when two iterations of 3-by-3 dilation of the proximal mask overlap the distal mask, giving a two-pixel Chebyshev tolerance. Foreground means alpha above 127; palette assignment requires Euclidean RGB distance below 60, and a color mask with fewer than four assigned pixels is treated as absent. Video-level TVR, LIE, and CPE are means over frames. On one deterministic 64^2 frame from each of the 4,020 released clips, threshold 60 as-

signs 96.4% of the evaluator’s alpha-gated foreground pixels and reaches 98.2% class accuracy among assigned pixels and 91.1% macro IoU against the released segmentation mapped to the evaluator classes. In a threshold sweep from 20 to 100, threshold 60 is within 0.1 percentage point of the best macro IoU; the $\alpha > 127$ evaluator foreground gate excludes 17.2% of segmentation-labeled foreground pixels, a foreground-definition scope difference rather than a classification error. Figure 4 shows one isolated trigger for each score. These are dataset-specific metrics that exploit the fixed limb colors: they query specific structural properties of the rendered figure, and none is a complete correctness judgement. For video, the evaluator adds color-mass drift, centroid speed and acceleration, motion fraction, angular speed and jerk, and height variation; some track the whole foreground, others individual limbs. Centroid speed is the mean foreground-centroid displacement in pixels per frame; motion fraction is the fraction of frame transitions whose centroid displacement exceeds the released 0.25-pixel threshold; angular jerk averages the absolute second differences of each limb’s principal-axis angle. These are image-space diagnostics, not biomechanics. The motion quantities are two-sided reference signals: zero can mean a stopped video, and large values can mean jitter. Agreement with the real reference, not a small value, is the target.

Occlusion produces non-zero topology scores even on real rendered frames, so scores on real rendered frames are a *real reference*, not an error floor. A deterministic manifest selects one camera per source motion—so correlated views of the same motion never enter as independent samples—and keeps the two 128-video reference halves motion-disjoint. For completeness, we report FVD [12] as a secondary coarse distributional reference, not a primary

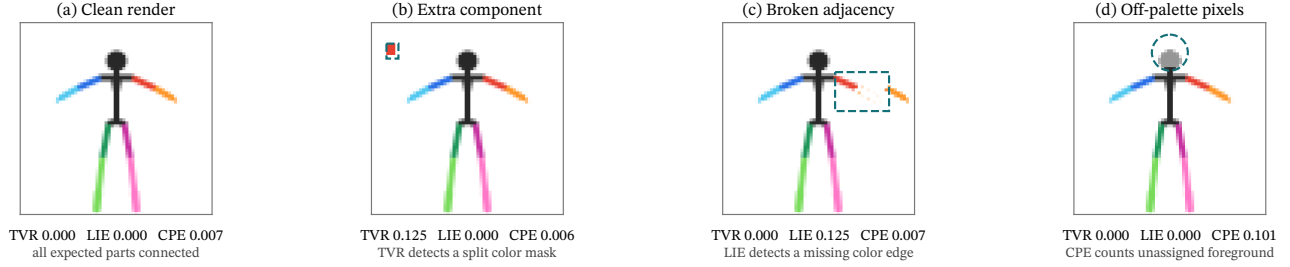


Figure 4. Controlled examples of the three structural scores at the released 64^2 evaluation resolution. All four panels derive from the same rendered frame and are scored by the released evaluator. A detached limb-color component raises TVR by $1/8$; translating one distal color mask beyond the two-pixel adjacency tolerance raises LIE by $1/8$; recoloring foreground pixels outside the palette raises CPE. The clean CPE is non-zero because some antialiased boundary pixels fall outside the RGB-distance tolerance. These edits illustrate the score definitions; they are not generated-video samples.

measure of video quality or a diagnostic score. We composite RGBA onto white, resize to 224^2 , and extract 400-dimensional features with the standard Kinetics-400 I3D TorchScript network. Because that network was trained on natural action videos and this study has only 128 held-out source motions, values are comparable only within this fixed implementation, window length, and sample count; close differences require uncertainty analysis.

Controlled validation. Five deterministic structural corruptions were applied to 500 held-out frames re-rendered from their stored labels, and scored against the same frames uncorrupted. A partial left/right limb swap raises LIE from .063 to .266; an extra arm raises TVR from .176 to .261. A whole-limb swap preserves the color graph and is not detected; bone stretching and hand deletion are also missed, because the metrics measure topology, not proportions or end-effectors. Temporal corruptions (freeze, shuffle, loop, reverse) are evaluated in §5.

5. Diagnostic validation and reference results

Metric stress test. Figure 5 reports the 64-frame study on protocol-matched first-64-frame windows ($n = 128$ per set). The real reference half B has centroid speed .373, centroid acceleration .415, and angular jerk .073. Freezing zeroes all motion signals. Shuffling raises the three signals to 1.802, 3.244, and .420; looping the first eight frames gives .336, .498, and .083. Reversal changes neither the frame set nor any of the time-symmetric hand-designed signals. FVD scores the corruptions against the 114.7 real-real reference at 620.8 for freezing, 520.7 for shuffling, 418.2 for looping, and 120.1 for reversal. Across 30 paired resamples of 64 videos from the 128-video source set, the paired FVD difference under reversal is 5.24 ± 5.24 (mean $\pm$ standard deviation; range -5.76 to 15.74), positive in 25/30 resamples. The hand-designed signals are reversal-invariant by

construction, and this FVD responds only weakly and inconsistently to reversal; neither certifies directionally correct motion.

Part-wise localization check. We applied a topology-preserving arm-chain freeze at four severities to 24 unflagged test motions, using camera 0 once per source motion. For each clip, the arm with greater clean rendered angular path was selected; its local bone directions were blended toward their first-frame values while attachment and bone lengths were preserved. The median reduction in the two affected color-part angular paths increased from 1.51 to 4.69, 8.61, and 9.11 rad across severities .25, .5, .75, and 1.0. Median absolute change across the other six limb parts remained .006–.011 rad; the affected reduction exceeded that change in 18/24, 23/24, 22/24, and 23/24 clips. At full severity, the 95% clip-bootstrap intervals are 4.58–16.43 rad for affected reduction and .001–.032 rad for unaffected absolute change. Median absolute TVR and LIE changes at full severity are .031 and .001, respectively: the part-wise signal localizes this arm-chain edit, while severe pose changes can still alter visible topology through occlusion.

I3D embedding sanity check. We validated the released FVD protocol at the distribution level on the final v0.2 release (camera 0 only, first 64 frames at the native frame rate, 134 prompts $\times$ 10 seeds), using ten deterministic matched- N partitions with $N=650$ videos per set. Splitting the same prompt roster by motion seed gives median FVD 26.37 (range 23.72–30.13), close to the random-video-half baseline of 24.61 (21.39–28.03), while disjoint group-balanced prompt rosters give 47.82 (39.51–57.32); the same-roster value was lower in 10/10 partitions. FVD therefore responds to dataset-level action-composition change and is relatively insensitive to seed variation, but it is not a per-video or prompt-correctness score. We also tested the

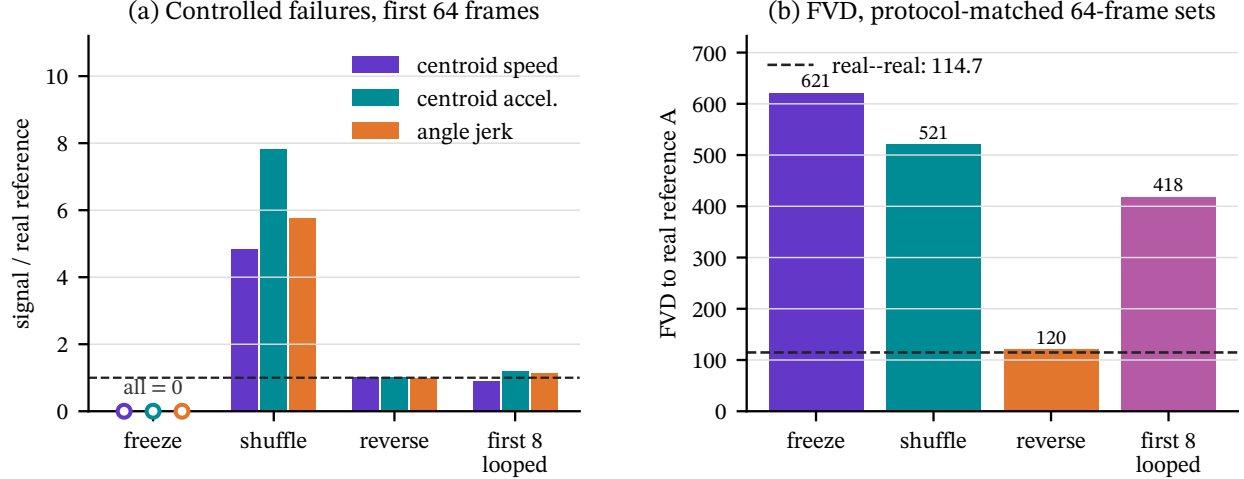


Figure 5. Controlled study on protocol-matched first-64-frame windows ($n = 128$ per set). *Freeze* repeats window frame 1, *shuffle* permutes the 64 selected frames, *reverse* reverses the order of those same 64 frames, *first 8 looped* repeats window frames 1–8. (a) Freezing zeroes the motion signals; shuffling and looping create excess acceleration and angular jerk; reversal changes neither. (b) Against the 114.7 real–real reference, FVD scores freezing 620.8, shuffling 520.7, reversal 120.1, and looping 418.2.

per-video I3D features directly, as pairwise Euclidean distances rather than FVD. Among 256 clean clips, 123 same-prompt, different-seed pairs have median distance 24.21 against 32.82 for matched different-prompt pairs within the same broad motion group; a same-prompt pair is closer in 80.4% of balanced comparisons, and nearest-neighbor prompt identity is 18.4% against a .38% random baseline. Paired clean-to-corruption median distances are 29.09 for freezing, 27.52 for shuffling, and 28.46 for looping, but only 9.34 for reversal: the embedding carries prompt-linked action information and reacts to several temporal corruptions, while reversal remains comparatively weak.

Training objective. Every released denoiser uses velocity-prediction flow matching on $x_t = (1 - t)x_0 + t\epsilon$, with $t = \text{sigmoid}(u)$, $u \sim \mathcal{N}(0, 1)$, and target $v^* = \epsilon - x_0$ (often called rectified flow, though we apply no re-flow step) [14, 15]. For the Image DiT, the loss is $\mathcal{L}_{\text{pixel}} = \mathbb{E}[(1 + \alpha)\|\hat{v} - v^*\|_2^2] / \mathbb{E}[1 + \alpha]$, where the ground-truth soft alpha is shared across the four RGBA channels. The three pixel-video DiTs in Table 3 add an RGBA clean-prediction auxiliary of weight 1. From $\hat{x}_0 = x_t - t\hat{v}$, it penalizes alpha disagreement, foreground RGB squared error, and a 0.1-weighted background RGB error. The term is gated by $(1 - t)$ and uses $w = 1 + 9 \text{clip}(a + |a - \hat{a}|, 0, 1)$ to emphasize visible and mispredicted foreground. Mini-Wan applies the corresponding observable-space terms after its frozen codec; because its primary loss is latent-space flow matching, its objective is analogous but not mathematically identical. Each model uses a frozen T5-small encoder attached through cross-attention. These rows are reference

points for users of the dataset.

- **Image DiT.** The 39.9M-parameter factorized backbone (4×4 frame patches, alternating spatial and temporal attention blocks) run with $T=1$; its checkpoint supplies the spatial and text weights for image initialization, an established practice [16, 17].
- **Pixel video DiTs.** The same 39.9M factorized backbone with $T=64$. The video rows separate random versus image-checkpoint initialization and, among image-initialized models, factorized attention versus a local $3 \times 3 \times 3$ convolutional mixer (a zero-initialized depthwise-plus-pointwise 3D convolution per block, about 1.9M extra parameters).
- **Dancing Stick Figures Mini-Wan 40M.** Our strongest released video reference has 39.4M trainable denoiser parameters and uses the Diffusers `WanTransformer3DModel`, retaining its Wan block stack: full self-attention over latent video tokens with 3D rotary position encoding and text cross-attention [19]. The scale and interface hyperparameters are adapted: width 384, 16 layers, 6 heads, (1, 2, 2) patches, and 8 latent channels on a $16 \times 16 \times 16$ grid, with frozen T5-small conditioning. It is trained from random weights and uses no pretrained Wan weights. For its latent velocity MSE, an alpha-derived occupancy $a_\ell \in [0, 1]$ gives each latent cell weight $1 + a_\ell$, normalized by the mean weight. An auxiliary term of weight 1 decodes both the predicted clean latent and the target latent through a separately trained frozen video codec. Let a and $\hat{a}$ be their decoded alpha values and $w = 1 + 9 \text{clip}(a + |a - \hat{a}|, 0, 1)$. The term, gated by $(1 - t)$, adds mean-normalized $w|a - \hat{a}|$,

squared RGB disagreement weighted by $wa\hat{a}$, and a 0.1-weighted background RGB penalty. The codec uses fourfold temporal and spatial compression with eight latent channels (f4t4d8). This is a compact Wan-architecture reference, not a reproduction of Wan2.1 at its published scale, codec, text encoder, data, or training recipe. We release the 30k checkpoint as the primary latent-space reference.

- **Codec control.** Real windows encoded and decoded by that same f4t4d8 codec, separating codec cost from model skill.

Optimization and conditioning mixture. All learned rows use AdamW, a 2×10^{-4} peak learning rate after 1,000 warm-up updates, cosine decay, and 10% text-conditioning dropout. The image stage uses batches of 128; the video stages use effective batches of 16. Each learned video row runs for 30,000 optimizer updates, or 480,000 sampled examples and 37.3 passes through the training loader. Pixel-video training draws a single-frame batch with probability .1; the remaining batches are text-to-video, with no first-frame input. Mini-Wan likewise draws a single latent-time batch with probability .1 and has no image-conditioning input.

An audit of the frozen f4t4d8 codec bounds its role: held-out foreground reconstruction MSE is 9.5% higher than training-prompt MSE, and neutralising background latent cells raises foreground error $5.5\times$. The mini-Wan row therefore includes both generator and codec error, and it cannot support a general pixel-versus-latent claim.

Table 3 collects the runs. Each block names its protocol, because single-frame and 64-frame scores are different measurements. Earlier v0.1 baselines used a superseded split and are not comparable rows. Figure 1 shows the 30k-step image-initialized factorized Pixel DiT.

Reading the comparisons. At matched exposure, image initialization improves the factorized Pixel DiT from TVR .259 to .161 and FVD 483.7 to 319.0. The three Pixel rows have speed .350–.366, close to the real-reference .373, while their motion fractions (.431–.470) remain below the real .501. The local mixer gives the lowest Pixel TVR (.149), jerk (.146), and FVD (282.9); the plain image-initialized factorized model gives lower LIE (.033) and CPE (.028), so neither dominates every diagnostic. Mini-Wan reaches FVD 182.6 but changes representation, backbone, and exact auxiliary construction, and its LIE and CPE are higher. Each learned row is one training run, so these are observed outcomes rather than estimates of architecture effects. Figure 1 shows the corresponding 30k-step image-initialized checkpoint across locomotion, gesture, transition, and dance prompts.

Why pixel space. For this release, at 32^2 – 64^2 , pixel-space training is practical and keeps thin-limb errors in the representation the evaluator scores. A separately trained video VAE adds a training stage and, as the audit of this released codec shows, can confound generator errors with codec artifacts. The core Colab therefore follows the pixel-space route.

6. Accessibility and reproducibility

The repository releases the dataset-generation harness, cache builder, trainers, sampling code, evaluator, corruptions, FVD evaluation, checkpoints, and comparison script. The Colab notebook is a reduced-budget reproduction of the reference two-stage training pipeline, not a reproduction of the released checkpoints’ quality: with the same code and architecture it builds a cache, trains the image generator for 2k steps, transfers its spatial weights, trains the video generator for 2k steps, samples a multi-second GIF, and compares scores against real references. The Table 3 reference checkpoints use 30k updates per stage. 32^2 is a derived smoke-test view; 64^2 is the reference dataset tier and model resolution. Each stage leaves an artifact before the next begins.

The motion release supports rebuilding the visual records from the public motion table: the rebuild script recomputes body and camera settings from each `clip_id` and reruns the public renderer, without ARDY. It rebuilds rendered frames from released motions; it does not regenerate the source motions. The verifier was run over all 514,800 128^2 frames of the pre-curation generation—a superset containing every one of the 482,400 released frames: every motion and metadata label matches; color and segmentation pixels agree on 99.88% and 99.92% of comparisons; joint visibility, the least stable field, agrees on 99.50%; depth and normals exceed 99.99%. A 360-frame check against the released 64^2 tier passes its declared agreement thresholds. The same script can rerender with seeded overrides to body scale, stroke, or camera, recorded in the manifest.

Reference runs were measured on one NVIDIA RTX PRO 6000 Blackwell (96 GB). The 30k-step image stage ran at 0.10 s/update with a 10.5 GB peak. The final factorized image- and random-initialized video runs each ran at 0.64 s/update and peaked at 37.8 and 37.7 GB; the local-mixer run took 1.24 s/update and peaked at 44.1 GB. We did not test the same configurations on lower-memory GPUs. The runs used different optimizer kernels, so these measurements document run cost rather than a controlled architecture-speed comparison and transfer only approximately to other hardware.

Our code is MIT and the dataset is CC0-1.0. Motions were generated with ARDY’s 20-fps Core model. ARDY’s code is Apache-2.0; its checkpoints are distributed under the

Table 3. Reference comparisons on the released seed-disjoint data. The three Pixel DiTs use the same training data, 30k optimizer updates, effective batch 16 (480k sampled examples; 37.3 loader epochs), optimizer and conditioning mixture, and alpha-weighted pixel-flow objective plus the RGBA auxiliary in the text. “Image” initialization copies compatible spatial and text weights from the single-frame checkpoint. Mini-Wan uses the same exposure but a different representation, backbone, and exact auxiliary construction, so it is separated as a latent-space reference. The codec control involves no training. Params count trainable denoiser parameters and exclude the frozen text encoder and, for Mini-Wan, the separately trained codec.

Model	Params	Attention	Init.	TVR↓	LIE↓	CPE↓	Speed	Motion frac.	Jerk	FVD↓
<i>Single-frame initializer: prompt-conditioned, $n = 512$, 50 sampling steps</i>										
Real validation frames	—	—	—	.113	.092	.039	—	—	—	—
Image DiT, 30k	39.9M	factorized	random	.128	.051	.030	—	—	—	—
<i>64-frame controls: 128 seed-disjoint source animations; learned rows use three sampling seeds</i>										
Real reference windows	—	—	—	.116	.093	.037	.373	.501	.073	114.7 / ref.
Codec reconstruction floor	—	—	—	.153	.118	.053	.383	.507	.106	127.1
<i>Loss-matched Pixel DiTs: 30k updates, 480k examples, 37.3 loader epochs</i>										
Video DiT, 30k	39.9M	factorized	random	.259	.042	.032	.366	.431	.172	483.7
Video DiT, 30k	39.9M	factorized	image	.161	.033	.028	.350	.447	.151	319.0
Video DiT + local mixer, 30k	41.8M	factorized + local mixer	image	.149	.050	.035	.355	.470	.146	282.9
<i>Latent reference: 30k updates, 480k examples, 37.3 loader epochs</i>										
Mini-Wan 40M, 30k	39.4M	Wan full 3D	random	.199	.101	.058	.423	.534	.148	182.6

TVR is topology violation rate, LIE limb-identity error, CPE color-purity error. Speed, motion fraction, and jerk are two-sided signals: agreement with the real row is the target. The codec-floor row passes the real windows through the frozen f4t4d8 codec used by the Mini-Wan rows; its score should be read against them. FVD is included only as a secondary coarse distributional reference. The real control row is reference half B; learned-row FVD compares generated videos with reference half A, and the codec control reconstructs half B and compares it with half A. The two halves are source-motion-disjoint. The 128 source animations are the distinct reference units, not the total number of generated videos. Each learned video row contains 384 outputs—one generation for each reference prompt under each of three sampling seeds. The structural and temporal scores pool those outputs; FVD is computed separately for each 128-video seed set and then averaged. With $n = 128$ and I3D features trained on Kinetics-400, it is not a calibrated stick-figure quality score and should not by itself rank nearby rows. Real training-split windows score FVD 125.5 against the same reference, which is also what a training-set replay would score, so models near this band need a memorisation check. All table and diagnostic FVD values directly composite premultiplied RGBA over white before I3D embedding.

NVIDIA Open Model Agreement, which permits derivative works and does not assert ownership of model outputs. The released frames are procedurally rendered and the released source motions are ARDY model outputs from text prompts; no directly captured human images or motion sequences are included in this release.

7. Limitations and future work

The domain is narrow by design: one raster style, one figure, orthographic cameras, no real-video proxy. The color-based evaluator measures visible topology, not bone length, joint angles, left/right semantics, or biomechanics; its measured blind spots include whole-limb swaps, proportion changes, and time reversal. The arm-chain localization study covers one topology-preserving motion edit; localization for other body parts and failure types remains untested. The motions inherit ARDY’s capabilities and biases; curation was prompt-level over inspected realisations from every seed, and we have not human-validated every remaining motion. The seed-disjoint test measures in-domain generation under unseen motion realisations of known prompts, not zero-shot prompts or human motion at large, and the color metrics do not establish prompt adherence. Each model row in Table 3 is a single training run evaluated with three sampling seeds; those seeds measure sampling variation, not training variance, and no row carries a scaling claim.

A learned rig estimator is future work that follows from these blind spots: mapping a generated video back to a rig and re-rendering it could expose rig-space failures the color metrics miss, such as unstable bone length or end-effector errors. Its scores would need validation on controlled malformations before use as evidence.

Broader impact. The release lowers the compute and data-access barrier to studying video generation and makes failures easier to inspect without collecting images of people. Its narrow synthetic domain is also a risk if results are presented as evidence about realistic human video: the motions inherit the source model’s biases, and the color-based diagnostics are not general video-quality or human-motion scores. We therefore document their measured blind spots, scope all claims to this dataset and protocol, and release the generator and evaluator so that reported results can be inspected rather than treated as opaque rankings.

8. Conclusion

Dancing Stick Figures provides a compact environment for training and evaluating small, domain-specific video-generation models end to end. Each rendered video is released with its source rig and generation parameters, together with the harness needed to generate additional data. The release also includes reference models and dataset-specific measurements of topology, part identity, color pu-

rity, and motion. These measurements are intended as diagnostic tools: they help identify what a model has learned, where its outputs break down, and whether a change in training or architecture improves a particular failure mode.

Together, these components support a complete, reproducible workflow for video-generation research. A researcher can generate or modify the data, train a 64-frame video model, inspect its outputs, measure specific failures, revise the model, and repeat the process at a manageable scale. The present dataset deliberately focuses on text-conditioned videos of a single rigged figure, but the same workflow can be extended with new motions, appearances, camera conditions, and diagnostics. By releasing the data, generation harness, reference models, and evaluator together, we aim to make video-generation research easier to enter, reproduce, and study through iteration.

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